IE 400 Final Project Paper
Group 9

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Felix Lu, Jiya Maheshwari, Advait Thakur

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1. Abstract

This experiment investigates how paper material influences the flight time of a paper airplane. Flight time, measured in seconds using dual stopwatches to reduce human timing error, serves as the primary response variable. Three materials: notebook paper (70 gsm), office paper (80–100 gsm), and cardstock (225 gsm) were selected as the experimental factor. To isolate the effect of material type, all other factors known to influence flight performance were held constant. These include wingspan, launch angle, temperature, and wind speed, each controlled through consistent measurement and by conducting the experiment in a closed indoor environment free of air disturbances. For each material type, ten flights were performed, with airplanes constructed to identical dimensions and launched using a standardized method to minimize variation in release technique. This design allows for a randomized comparison of flight times across materials while reducing confounding sources of variability. The results of this study aim to determine whether heavier or lighter paper materials yield longer flight durations, offering insight into how structural properties affect aerodynamic performance in simple glider designs.

2. Introduction

Paper airplanes provide a simple yet effective way to study fundamental principles of aerodynamics, including lift, drag, and stability. Although paper airplanes are often viewed as toys, their flight performance depends on the same physical forces that govern real aircraft. Small changes in structural properties such as weight, stiffness, and flexibility can meaningfully alter the way an airplane moves through the air (Du et al., 2020). Because paper airplanes are inexpensive to construct and easy to test, they serve as an ideal system for examining how material characteristics influence flight behavior. It is important to understand these relationships because they provide insight into basic design optimization issues encountered in engineering, where material selection directly affects performance outcomes.
The main purpose of this experiment is to determine how paper material type affects the flight time of a paper airplane. Flight time was selected as the response variable because it reflects the airplane's ability to maintain stable flight. A longer flight time could indicate improved performance depending on the material of the paper airplane. The material types studied differ in thickness, weight, and rigidity, which are characteristics that are expected to influence flight times. By isolating material type as the treatment factor and holding all other factors constant, we aim to identify which material leads to the longest and most consistent flight times.

To ensure that material type is the only factor affecting flight performance, several potential sources of variation were controlled. All airplanes were folded using identical dimensions and methods to prevent shape variability. Launch angle, launch method, and wingspan were kept constant, and all trials were conducted indoors to eliminate environmental disturbances such as wind. Ten flights were performed for each material type, allowing for a randomized comparison of flight times while minimizing measurement noise. Through this controlled experimental setup, the study seeks to evaluate whether heavier or lighter paper materials provide a performance advantage and to better understand the influence of material properties on flight behavior.

3. Experimental Design

The goal of this experiment is to analyse how paper material type affects the flight time of a paper airplane. Since we are holding all variables constant except for the paper material type, this is classified as a One Factor Completely Randomised Design.

A one factor completely randomised design is an experimental design where we have one independent variable (factor) with two or more levels, and experimental units are randomly assigned to each treatment level. The randomisation ensures that each experimental unit has an equal chance of receiving any treatment, which controls for confounding variables (like temperature, win speed etc. in our case) and allows us to make valid conclusions about the factor's effect on the response variable.

3.1 Mathematical Model

$$Yij = \mu + \tau i + \varepsilon ij$$

Where:
Yij = flight time of jth flight of a plane with ith material
μ= avg flight time across all materials
τi = effect of the ith material (1 = Notebook Paper, 2=Office Paper, 3=Cardstock)
εij = error term for the jth flight of a plane with ith material

We assume that the errors are independent and identically distributed N(0,**σ**^2) for ease of conducting the experiment.

3.2 Process to gather the data

The data collection process remained consistent across all trials in order to maintain randomization. We constructed 30 airplanes: 10 each with Notebook paper (70gsm), Office Paper(80-100gsm) and Cardstock (225gsm). We created these airplanes using the same design(Fig 1 shows the design we followed) and paper dimensions to keep uniformity. We labelled each of the 20 airplanes from 1-30 and shuffled them to maintain ambiguity. We recorded the material type for each plane prior to testing as well.

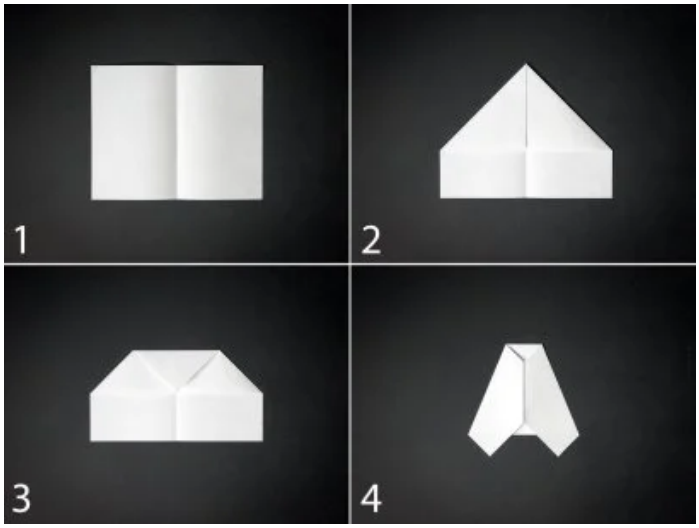

Fig 1: Simple Paper Airplane Design

In order to randomise the order of flights we used a random number generator. The airplanes were flown in that very order. This randomised the material type being tested. For each trial, the designated airplane was launched from a marked starting position in the apartment hallway. Two people operated stopwatches simultaneously to measure flight time from the moment the airplane left the thrower's hand till it touched the ground.

The flight times from both stopwatches were averaged to reduce human timing error. After each flight, the data was immediately recorded in a spreadsheet with columns for trial number, material type, stopwatch 1 time, stopwatch 2 time, and average flight time.

3.3 Response Variable

The response time is the time taken (in seconds) to leave the thrower's hand and touch the floor for the first time. The main reason to choose this response variable is to understand how material affects the stability of a flight.

3.4 Treatment Factor and Levels

The experimental factor is paper material type, which has three levels:

Level 1: Notebook Paper (70 gsm)

Level 2: Office Paper (80-100 gsm)

Level 3: Cardstock (225 gsm)

3.5 Error Prevention and Safety

The following guidelines were followed in order to prevent errors and maintain safety:

1. The experiment was conducted in the hallway during break (i.e. negligible footfall).

2. Only paper materials were used for experimentation, which do not pose any significant risk.
3. The airplanes were folded with the same design across materials.
4. The airplanes were numbered 1-30 and their types were noted.

3.6 Recording Data

All data was recorded on an excel spreadsheet immediately after each flight. The spreadsheet included columns for trial number, randomization order, material type, stopwatch 1 time, stopwatch 2 time, average flight time.

3.7 Randomization

All the planes were labelled with a marker with numbers from 1-30. As previously mentioned, we used a random number generator to generate a number between 1-30 which helped avoid clustering of airplanes with the same material type (fig 2 shows 1 out of the 10 office paper airplanes).

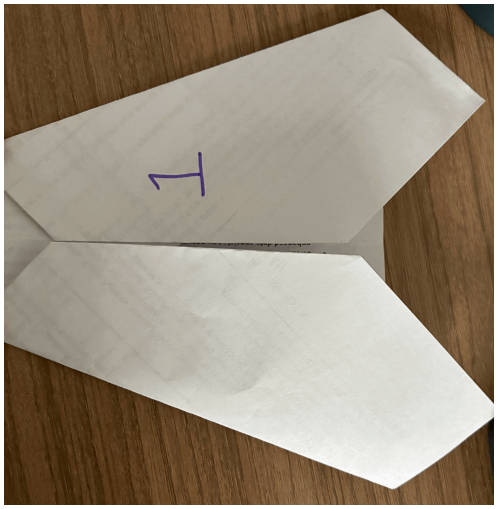

Fig 2: Office Paper airplane design with numbering

3.8 Assumptions for the Model

We use a one-factor completely randomized design in this experiment which assumes the following about the data:

A. *Independence:* Each flight time measurement is independent of all others
B. *Normality:* The error terms εij are normally distributed
C. *Constant Variance (Homoscedasticity):* The variability in flight times should be roughly the same for all three material types

These assumptions are necessary for the validity of the ANOVA F-test.

4. Methodology

In order to conduct a complete statistical analysis of our one factor randomized experiment, we will be using Analysis of Variance (ANOVA) as our key approach. Additionally, we will be utilizing confidence intervals and Tukey's as well as Scheffe's method in order to verify model assumptions and to identify differences between material types.

4.1 Analysis of Variance (ANOVA)

The main analysis will involve a one factor ANOVA, where the null and alternative hypotheses are:

H_0: $\mu 1 = \mu 2 = \mu 3$ (all 3 materials have the same mean flight time)

H_1: At least one material type has a different mean flight time

The ANOVA splits total variability into two categories. The first one being the treatment effect which is the difference between material types and random error is the variability within material types as per the mathematical model in Section 3.

We use the following formula to calculate the F statistic for ANOVA.

$$F = MStreatment \,/\, MSerror$$

We will reject the null hypothesis if the F-statistic exceeds the critical value from the F-distribution at significance level $\alpha = 0.05$. This means that the paper material type has a significant effect on flight time. The ANOVA table can be found in the next section(Table 3).

4.2 Model Assumptions

We will visualise 3 plots in order to verify if the errors meet our model assumptions of normality, homoscedasticity and independence.

1. **Residuals vs. Run Order Plot:** This plot shows the actual flight time minus the predicted flight time on the y axis and the run order on the x axis. This plot identifies if we meet the Independence assumption. This assumption is met if there is random scatter around 0.
2. **Residuals vs. Predicted Values Plot:** This plot displays residuals against the predicted flight times for each material. This checks if the equal variance or homoscedasticity assumption is met if the points scatter around 0 evenly and there's no obvious pattern.
3. **Q-Q Plot:** We normalise the residuals and plot it against a straight line graph to see whether the residuals are approximately normal.

If our model violates any of these assumptions, we will mention them in the limitations section.

4.3 Parameter Estimation

Treatment effects tell us the deviation of a material's mean flight time from the overall mean across different materials. We will be estimating these values for the 3 treatments effects for the different materials $\tau i\ hat$ i = 1,2,3. Based on these we can see which material has a longer or shorter flight time on average as compared to others.

We can also estimate the overall variance not described by the type of material using the MSE from the ANOVA table.

4.4 Confidence Intervals

We will construct 95% confidence intervals for the mean flight time of each material type. Narrower confidence intervals indicate more precise estimates, while wider intervals indicate variability in the data which may lead to uncertainty in our estimates.

4.5 Multiple Comparisons

ANOVA only tells us whether there is a significant difference between means of difference materials. To find which specific pairs of means have significant differences, we use Tukey's and Scheffe's method:

1. **Fisher's LSD:** We use Fisher's LSD to compare means for different materials two at a time but using overall variability.
2. **Tukey's Method:** We use Tukey's method to compare means for different materials two at a time.
3. **Scheffe's Method:** Allows us to test any difference between means of any material.

We will use these methods to confirm that our conclusions are statistically sound.

5. Results

The One-Factor Completely Randomized Design ANOVA was conducted to evaluate the effects of paper material on flight time. The experiment has 30 total trials, with 10 randomized trials for each of the three levels of material: Notebook Paper (Level 1), Office Paper (Level 2) and Cardstock (Level 3). Table 1 presents a summary of the descriptive statistics, showing the sample size N, sum, mean flight time, and variance estimates for each paper material type.

Level (i)	Material	Sample Size (n)	Sum (Yi.)	Mean (Ybar_i.)
1	Notebook Paper	10	37.1	3.71
2	Office Paper	10	31.8	3.18
3	Cardstock	10	24.6	2.46
Total	**Grand Total**	**N = 30**	**Y.. = 93.50**	**Ybar_.. = 3.12**

Table 1: Summary of descriptive statistics

Prior to performing Analysis of Variance, the data was initially visualized to identify trends in flight time across the three paper materials.

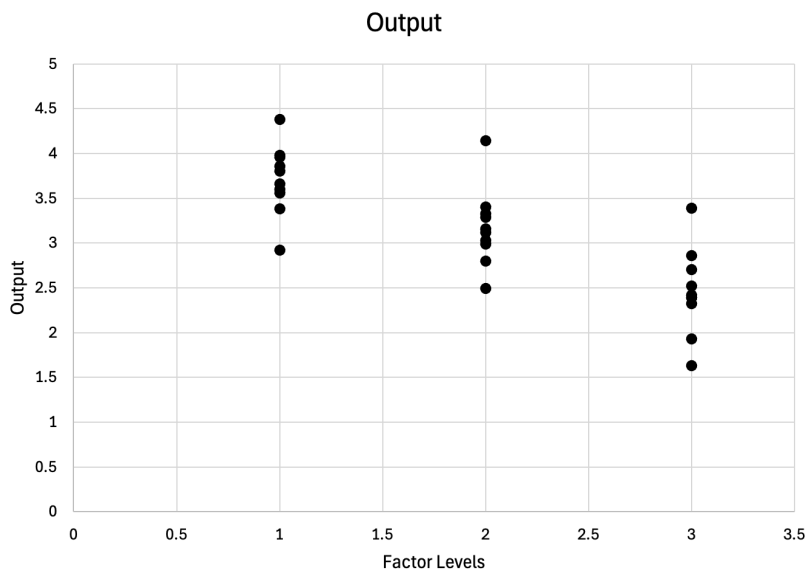

Fig 3: Plot for Output vs Factor Levels

The output (Flight time) vs Factor Levels (Different Paper Materials) as shown in figure 3, shows a distinct variation in mean flight time between the materials.

Confidence interval 95%		
Level	**Lower**	**Upper**
1	3.42637365	3.99362635
2	2.89137365	3.45862635
3	2.17137365	2.73862635

Table 2: 95% Confidence Intervals for individual treatment means.

Furthermore, the 95% confidence intervals for the individual treatment means were calculated to provide a baseline for expected performance at each level, as shown in Table 2.

Observation: These visualizations suggest that Notebook Paper (Level 1) yields the longest flight times, while other papers' yields are low comparatively.

5.1 Analysis of Variance (ANOVA)

The Analysis of Variance was performed to test the null Hypothesis (H0), stating there is no difference in mean flight times for the three different materials of paper used in the experiment.

- Null Hypothesis (H0) : mu(Notebook) = mu(Office Paper) = mu(Cardstock)
- Alternate Hypothesis (H1): At least one material mean is different

ANOVA Table						
Source	**Sum SQ**	**Dof**	**Mean SQ**	**F0**	**P-value**	
Treatment	7.93216667	2	3.96608333	20.7563819	3.47133E-06	Significant
Error	5.1591	27	0.19107778			
Total	13.0912667	29				

Table 3: Analysis of Variance (ANOVA) results for flight time.

The results of the analysis are summarized in Table 3. The **F-statistic = of 20.76** and **P-value = 3.47*10^-6** were obtained from the analysis. At **alpha = 0.05**, we have significant evidence against the null hypothesis. Thus, we reject the null hypothesis, and conclude that paper material has a significant impact on the flight time of a paper plane.

5.2 Model Adequacy Checking

To ensure validity of the ANOVA results found in Table 3, following core assumptions regarding Error were checked:

1. **Assumption of Independence:**

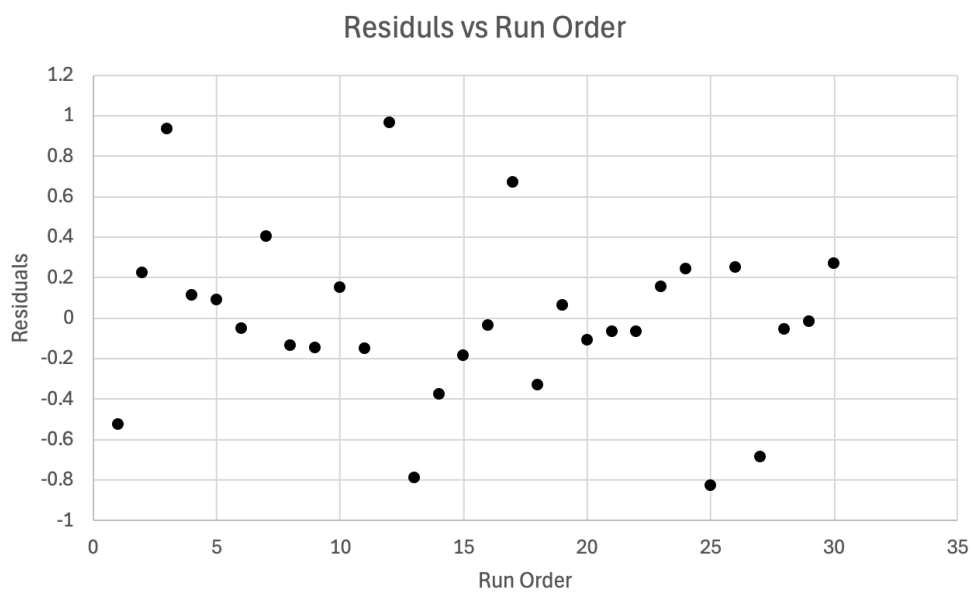

Figure 4: Plot of Residuals vs. Run Order.

The plot of Residuals vs Run Order (Figure 4)displays a random scatter with no discernable pattern or trend. This indicates that the data was gathered in a random order and there are no time-dependent variables, such as thrower fatigue or environmental changes) influencing the results.

2. **Assumption of Constant Variance:**

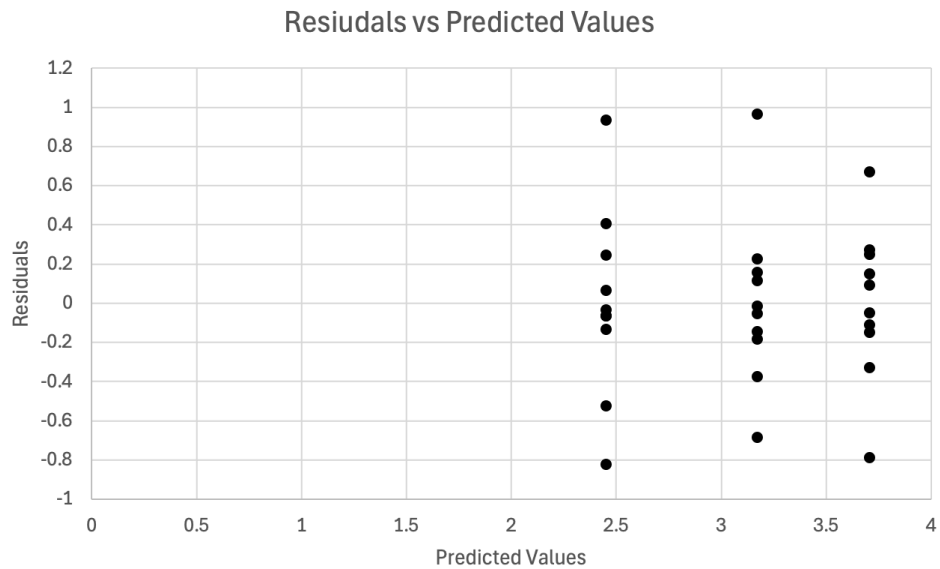

Figure 5: Plot of Residuals vs. Predicted (Fitted) Values.

The plot of Residuals vs Predicted values (Figure 5) shows relatively consistent variance (spread) across all three material groups. The lighter Notebook paper has slightly higher variability due to its tendency to flutter, the difference is high enough to violate this assumption.

3. **Assumption of Normality:**

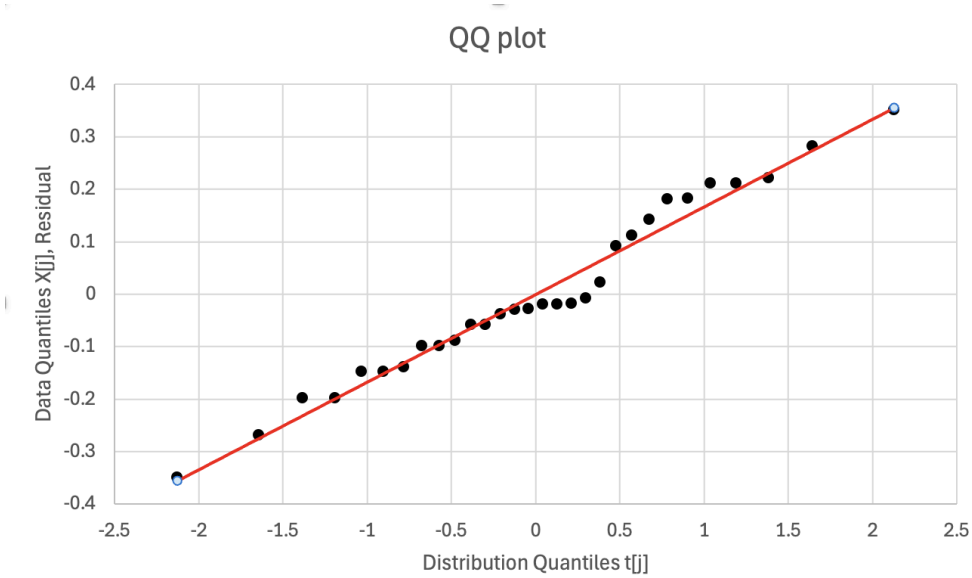

Figure 6: Normal Probability Plot (QQ Plot) of Residuals

The QQ plot (Figure 6) of the residuals obtained shows that all the data points fall approximately along a straight line. This indicates that the normality assumptions are valid for the F test.

5.3 Parameter Estimation

As ANOVA indicated significance and the model assumptions were satisfied, the model parameters were estimated to quantify the effect of each paper type. These estimates are presented in Table 4.

The estimated coefficients represent the deviation of each treatment mean from the grand mean: $\hat{\tau}_i = \bar{y}_{i.} - \hat{\mu}$

Parameter Estimate	
Mu_hat	3.113333333
tau1_hat	0.596666667
tau2_hat	0.061666667
tau3_hat	-0.658333333
sigma^2_hat	0.191077778

Table 4: Model parameter estimates.

The Mean Squared Error was utilised as an unbiased estimator of the error variance.

5.4 Multiple Comparisons

To determine exactly which pairs of paper materials are significantly different, pairwise comparisons were performed. Three methods were used to construct confidence intervals for the differences in means: Fisher's LSD (Normal), Tukey's HSD, and Scheffé's method. The results are summarized in Table 5.

95% CI on Differences in Treatment Means						
	mu1 - mu2		mu1 - mu3		mu2 - mu3	
Yi._bar - Yj._bar	0.535		1.255		0.72	
	Lower	Upper	Lower	Upper	Lower	Upper
Individual	0.13389177	0.9361082	0.8538918	1.65610823	0.31889177	1.12110823
simult, Tuckey's	0.049809591	1.0201904	0.7698096	1.740190409	0.23480959	1.205190409
simult, Sheffe's	0.028679996	1.04132	0.74868	1.761320004	0.21368	1.226320004

Table 5: Pairwise comparisons of paper materials using Fisher's LSD, Tukey's HSD, and Scheffe's method.

5.5 Interpretation

The pairwise comparisons reveal the following:

1. Best Performer: Notebook Paper (Level 1), it produced the longest flight times. It significantly outperforms Office Paper by an average of 0.535 seconds and Cardstock Paper by 1.255 seconds
2. Average Performer: Office paper (Level 2), it performed significantly worse than the Notebook Paper but performed much better than the Cardstock Paper by 0.72 seconds.
3. Worst Performer: Cardstock (Level 3), it resulted with the shortest flight times amongst the three.

As shown in Table 5, the intervals for all pairs (e.g., Notebook vs. Cardstock) do not contain zero across all three methods, confirming a statistically significant difference in performance between every material tested.

5.6 Statistical Significance

All three methods confirm that every pair of materials is significantly different. Since there are no zeros in any of the intervals, there is strong evidence that the difference in flight times are real and not due to random chance.

Fischer's LSD (individual Confidence interval) produced the narrowest intervals whereas the Sheffe's method produced a wider interval for the pairwise differences. Even under the widest confidence intervals, the lower bounds remained positive for all differences. This confirms the results of Notebook Paper being the best performers amongst the three.

6. Discussion and Limitations

6.1 Discussion

The results of this experiment suggest that paper type does have an effect on the flight time of a paper airplane. As confirmed in the results section, the notebook paper seemed to be the best performer of the three materials, followed by the office paper and then the cardstock. The differences in flight time between notebook paper, office paper, and cardstock appear to be related to how each material handles weight and stiffness. Lighter papers tend to stay in the air longer because they create less downward force, while heavier or stiffer papers such as the cardstock paper will fall faster even though they provide more structure. These observations support the idea that material properties play an important role in overall flight performance. By keeping all other factors constant, it was easier to see the influence of paper type alone.

The experiment thus confirmed that changes in the weight and density of the material can lead to noticeable changes in flight behavior. Despite all the materials being paper-based, the flight time of the plane was still significantly altered.

Another important point from the experiment is how difficult it is to control all variables across the experiment. Even though folds were standardized and the launch method was kept the same, small variations still occurred. However, the results still showed clear trends across materials, indicating that the material effect was strong enough to be noticeable despite natural human variation. This strengthens confidence in the conclusion that paper type does affect flight time.

It was observed that the three paper materials all had very similar variability. This could be due to the controlled nature of the experiment, where many feasible conditions were kept constant, such as the indoor setting, launch movement, and plane design. The residual plots asserted the core ANOVA assumptions were reasonably satisfied, so the reliability of the experiment was supported.

Beyond the scope of this experiment, the results that we found can apply to real world situations, where material weight and density is an important attribute of materials that are used in everyday scenarios. Choosing the right materials for the construction of something can significantly change the project's stability, strength, and durabliity. When engineers construct any sort of project, it is essential to consider the material's various physical properties in order to make the most well-informed decisions.

6.2 Limitations

Throughout the experiment, we noticed that there were a few limitations that affected the strength and generalizability of this experiment. One major limitation is that all launches were performed manually by person. This results in human error as we cannot precisely control our own bodies similar to a machine. Even with consistent technique and practice, it is impossible for a person to reproduce the same release angle, release speed, and wrist motion perfectly each time. Small differences in the throw can influence how high the airplane starts, how quickly it levels out, and how long it stays airborne. Using a mechanical launcher or a guided release mechanism would have greatly reduced this type of variation.

Another limitation of the experiment is that we only used one design for all of the paper airplanes. It is certainly possible that airplanes with larger wings might benefit more from lighter paper, while airplanes with pointed noses might require stiffer paper to maintain structural integrity during flight. Because of this, we cannot necessarily conclude that the results of the experiment would apply to all kinds of paper airplanes, but only for the design that we had chosen for the experiment.

The folding process itself introduces additional variability. Even when following identical steps, small differences in crease alignment, fold tightness, and wing symmetry can occur. These imperfections may cause an airplane to tilt or drift, change its lift force, or lose stability sooner. Using templates, rulers, or creating a machine that could fold paper airplanes could make the folding process more consistent, reducing variability caused by human error.

The different environmental factors also present limitations. Even though the experiment was conducted indoors in an apartment to eliminate wind, these apartment hallways can still have slight air currents from ventilation systems, open doors, or movement in the hall. Air pressure and humidity can also vary depending on the day and time of testing. While these effects are small, they can still influence the flight of a lightweight paper airplane, especially when differences between materials are subtle. A paper airplane may become slightly damper due to humidity, or the air conditioning may blow a paper airplane of course. These are environmental limitations that can hamper the results of the experiment.

Finally, another limitation is that this experiment was conducted in the winter time. Each season of the year presents different atmospheric conditions, whether it be through air pressure, temperature, etc. Since it is impossible to regulate the pressure in an apartment complex by ourselves, these factors could affect different planes differently depending on the material used. Since this variable was not tested during the experiment, it is unknown whether or not it has a significant effect on the experiment.

7. Conclusion

The primary objective of this experiment was to determine the effect of paper material on flight times for a handcrafted model. Three different paper materials were selected for this experiment, Notebook Paper (70 gsm), Office Paper (80-100gsm) and Cardstock Paper (225 gsm). The experiment was successful in isolating the influence of material properties on aerodynamic performances.

The Analysis of Variance (ANOVA) results in table 3, providing strong evidence to reject the null Hypothesis (H0 = Flight times are unaffected by the material of the paper). The F statistic (20.76) and P-value (<0.001) were obtained from ANOVA as evidence for our conclusion, confirming that paper material significantly impacts the flight duration.

Post-hoc analysis using Fisher's LSD, Tukey's test and Scheffe's test (Table 5) supported the ANOVA results and produced a clear hierarchy amongst the three paper materials:

- Notebook Paper yielded the longest average flight time, 3.71 seconds
- Office Paper yielded an average flight time compared to the others, 3.18 seconds
- Cardstock Paper yielded the shortest flight time, 2.46 seconds

The data visualized in Figure 1 demonstrated an inverse relationship between the weight of the paper and flight time. The lighter Notebook Paper had a slower decent rate and extended loft time. On the other hand, the Cardstock paper had a greater structural rigidity, but the heavy weight increased the descent acceleration, overriding the stability benefits and resulting in significantly shorter flight times.

The model's validity and error assumptions were verified through independence, constant variance and normality checks as shown in Figures 4, 5 and 6. Although several uncontrolled environmental factors resulted in perturbed individual observations, the magnitude of treatment effect was robust. The results yielded were consistent, statistically significant differences across pairwise comparisons.

Based on the experimental results, minimizing the weight of the paper was a more critical factor in maximising flight times than increasing material stiffness. Future experiments can be carried out to determine the interaction between weight of the paper and different geometry to determine if heavier materials offer better results in high-speed and outdoor conditions.

8. References

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